

LIRA - Use Case Full Description

Use Case Full Description: Ask AI Chatbot

Use Case Name	Ask AI Chatbot
Scenario	A patron queries the AI for research assistance.
Triggering Event	Student or Faculty opens the LIRA chat interface and submits a question.
Brief Description	Enables users to interact with an AI assistant to get immediate answers, navigate library services, or seek research guidance.
Actors	Student/Faculty (Primary), AI Processing Service (Secondary)
Related Use Cases	Extends: Generate AI Response
Stakeholders	Library Patrons, Library Administration
Preconditions	User is accessing the LIRA portal and the AI Processing Service is operational.
Postconditions	The user receives a relevant, AI-generated response to their query.

Flow of Activities

Actor	System
1. Opens chat interface and types a question/request.	1. Initializes chat session and displays typing indicator.
2. Submits the query.	2. Receives query and routes it to the AI Processing Service.
3. Waits for response.	3. Analyzes intent, generates contextual answer, and displays it to the user.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
AI Service Timeout / Unreachable	System halts typing indicator and displays a graceful fallback message.
Inappropriate or Out-of-Scope Query	System identifies policy violation and generates a courteous refusal response.

Use Case Full Description: Perform Search

Use Case Name	Perform Search
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Scenario	A patron searches for specific books or research topics.
Triggering Event	User inputs keywords into the main search bar.
Brief Description	Allows users to query the library's database and utilizes AI to suggest related materials.
Actors	Student/Faculty (Primary), AI Processing Service (Secondary)
Related Use Cases	Includes: View Library Catalog; Extends: Generate Personalized Recommendation
Stakeholders	Library Patrons, Capstone Development Team
Preconditions	The Koha ILS catalog database is accessible.
Postconditions	User is presented with matching resources and recommendations.

Flow of Activities

Actor	System
1. Enters search terms and applies desired filters.	1. Validates input and queries the underlying catalog database.
2. Reviews retrieved list of catalog items.	2. Displays catalog items and triggers the Recommendation Engine.
3. Explores suggested related materials.	3. Displays personalized recommendations alongside standard search results.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
No matching records found in catalog	System displays a 'No materials found' message and suggests broader terms.

Use Case Full Description: View E-Book Summaries

Use Case Name	View E-Book Summaries
Scenario	A patron requests a quick overview of a digital resource.
Triggering Event	User clicks the 'Generate Summary' button on an e-book's detail page.
Brief Description	Provides users with an AI-synthesized summary of e-books or research papers.
Actors	Student/Faculty (Primary), AI Processing Service (Secondary)
Related Use Cases	Includes: View Library Catalog; Extends: Generate E-Book Summaries
Stakeholders	Students, Researchers, Library Administration

Preconditions	Target e-book has a valid digital text file available for parsing.
Postconditions	A concise, accurate summary is displayed on the user's screen.

Flow of Activities

Actor	System
1. Navigates to a specific e-book record.	1. Displays the book details from the database.
2. Clicks 'Generate Summary'.	2. Sends the text/metadata to the AI Processing Service for synthesis.
3. Reads the generated summary.	3. Renders the synthesized overview on the interface.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Text file is DRM-protected or unreadable	System alerts the user that a summary cannot be generated for this format.

Use Case Full Description: Request Citation

Use Case Name	Request Citation
Scenario	A patron requires an academic citation for a selected resource.
Triggering Event	User selects a material and requests a specific citation format.
Brief Description	Automates the generation of academic citations for books, journals, and e-books.
Actors	Student/Faculty (Primary), AI Processing Service (Secondary)
Related Use Cases	Extends: Generate Citation
Stakeholders	Students, Faculty, Capstone Development Team
Preconditions	Selected material has metadata recorded in the system.
Postconditions	A properly formatted citation string is provided to the user.

Flow of Activities

Actor	System
1. Selects 'Cite this' on a catalog item.	1. Prompts user to select desired citation format (APA, IEEE, MLA).
2. Selects the format.	2. Extracts catalog metadata and routes to the Citation Generator.

3. Copies generated citation to clipboard.	3. Outputs the strictly formatted academic citation.
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Exception Conditions

Exception Condition / Trigger	System Handling / Response
Incomplete source metadata	System prompts user to manually input missing fields before finalizing.

Use Case Full Description: Manage Library Catalog

Use Case Name	Manage Library Catalog
Scenario	A librarian updates the core repository of library materials.
Triggering Event	Librarian initiates an add, edit, or delete action in the catalog dashboard.
Brief Description	Central administrative function for maintaining library records and triggering immediate AI indexing.
Actors	Librarian (Primary), Koha ILS (Secondary), AI Processing Service (Secondary)
Related Use Cases	Extends: Generate Citation, Generate E-Book Summaries
Stakeholders	Library Staff, Koha Administrators
Preconditions	Librarian is logged in with administrative privileges.
Postconditions	Catalog database is updated, and the new record's summary and citation index are generated immediately.

Flow of Activities

Actor	System
1. Selects 'Add New Resource' and inputs metadata.	1. Validates input data against cataloging standards.
2. Saves the new record.	2. Commits the record to the Koha ILS database.
3. Configures auto-generate settings.	3. Sends the new record to the AI Processing Service, which immediately generates its summary and citation index and stores them in the Vector Database.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Koha ILS synchronization failure	System saves changes locally in a queue and alerts the librarian.

Use Case Full Description: Manage Book Loans

Use Case Name	Manage Book Loans
Scenario	Librarian processes the borrowing or returning of a physical book.
Triggering Event	Patron presents a book for checkout or return, or daily cron job evaluates due dates.
Brief Description	Handles the lifecycle of physical asset borrowing and overdue notifications.
Actors	Librarian (Primary), Koha ILS (Secondary)
Related Use Cases	Extends: Send Overdue Alert
Stakeholders	Library Staff, Patrons
Preconditions	Patron has a valid library account with no active blocks.
Postconditions	Item's status is updated to 'Borrowed' or 'Available' in Koha ILS.

Flow of Activities

Actor	System
1. Scans patron ID and book barcode.	1. Retrieves patron record from Koha ILS and updates book status to borrowed.
2. Confirms due date with the patron.	2. Calculates due date and registers the loan in Koha ILS's circulation database.
3. Monitors loan duration.	3. Triggers automated overdue notification if due date passes.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Patron reached maximum borrowing limit	System rejects transaction and displays a notification.

Use Case Full Description: Manage FAQ

Use Case Name	Manage FAQ
Scenario	Librarian updates the knowledge base for common user inquiries.
Triggering Event	Librarian accesses the FAQ administration module.
Brief Description	Allows library staff to maintain frequently asked questions assisted by chatbot interactions.
Actors	Librarian (Primary), AI Processing Service (Secondary)

Related Use Cases	Extends: Generate FAQ
Stakeholders	Library Staff, Patrons
Preconditions	Librarian has content management permissions.
Postconditions	Public FAQ page and chatbot knowledge base are updated.

Flow of Activities

Actor	System
1. Requests analysis of recent common queries.	1. Aggregates chat logs and generates suggested questions/answers.
2. Reviews and edits AI-suggested FAQs.	2. Saves modifications to draft database.
3. Publishes new FAQ entries.	3. Updates public portal and synchronizes the AI Processing Service's knowledge base.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Insufficient chat data	System informs librarian that manual FAQ creation is required.

Use Case Full Description: Manage Chatbot

Use Case Name	Manage Chatbot
Scenario	Librarian configures the behavior and parameters of the AI assistant.
Triggering Event	Librarian navigates to the Chatbot Configuration dashboard.
Brief Description	Administrative interface to tune AI model prompt parameters, tone, and knowledge sources.
Actors	Librarian (Primary), AI Processing Service (Secondary)
Related Use Cases	None
Stakeholders	Library Administration, Capstone Team
Preconditions	Librarian has system admin privileges.
Postconditions	Chatbot operational parameters are updated.

Flow of Activities

Actor	System
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1. Adjusts chatbot operational parameters.	1. Validates new configuration settings.
2. Tests new configuration in sandbox environment.	2. Applies settings to a temporary instance and processes test queries.
3. Deploys configuration to production.	3. Restarts the production Chatbot Synchronizer & Manager component with updated parameters.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Invalid parameter configuration	System prevents deployment and highlights conflicting settings.

Use Case Full Description: Manage Reports

Use Case Name	Manage Reports
Scenario	Librarian requests operational analytics on library usage.
Triggering Event	Librarian selects a date range and report type in the Analytics dashboard.
Brief Description	Compiles data on catalog searches, loan frequency, and AI utilization.
Actors	Librarian (Primary)
Related Use Cases	Extends: Generate Reports
Stakeholders	Library Administration
Preconditions	Analytics tracking database is populated with historical data.
Postconditions	Compiled analytical report is presented or downloaded.

Flow of Activities

Actor	System
1. Selects reporting metrics and timeframe.	1. Validates parameters and queries analytics database.
2. Clicks 'Generate Report'.	2. Compiles data into visualizations and summaries.
3. Exports report as PDF or CSV.	3. Formats data and triggers file download.

Exception Conditions

Exception Condition / Trigger	System Handling / Response
Requested date range contains no data	System displays an empty state message advising user to adjust timeframe.

